Data Cleaning Steps:

Sales Dataset

1.Understand the Dataset:

Identify columns, data types, and meaning

sales.head()

sales.tail()

sales.info()

sales.describe()

sales.dtypes

sales.columns

2.Standardize column names:

Clean inconsistent values

- Text case (lower / upper)
- Trim spaces
- Fix spelling issues

sales.columns = sales.columns.str.lower().str.replace(' ', '_')

3. **Removing duplicates**

sales.drop_duplicates(inplace=True)

4.Handle Missing Values

#replacing null values

sales.isnull().sum():Checking null values

Fill values

- Numerical → mean / median / 0
- Categorical → mode / “Unknown”/-

sales['store_id']=sales['store_id'].fillna('-')

Customers Dataset

1.Understand the Dataset:

customers.head()

customers.tail()

customers.info()

customers.describe()

customers.dtypes

customers.columns

2.Standardize column names

customers.columns = customers.columns.str.lower().str.replace(' ', '_')

3.removing duplicates

customers.drop_duplicates(inplace=True)

4. Correct data types

customers['age'] = pd.to_numeric(customers['age'], errors='coerce').astype('Int64')

5.handling missing values

#customers.isnull().sum():checking null values

filling missing values with median

customers['age'] = customers['age'].fillna(customers['age'].median())

Products Dataset

1.Understand the Dataset:

products.head()

products.tail()

products.info()

products.describe()

products.dtypes

products.columns

2.Standardize column names

products.columns = products.columns.str.lower().str.replace(' ', '_')

3.Remove duplicate rows

products.drop_duplicates(inplace=True)

4.handling missing values

products.isnull().sum():checking null values

filling missing values with unknown

products['brand']=products['brand'].fillna('unknown')

Returns dataset

1.Understand the Dataset:

returns.head()

returns.tail()

returns.info()

returns.describe()

returns.dtypes

returns.columns

2.Standardize column names

returns.columns = returns.columns.str.lower().str.replace(' ', '_')

3.Remove duplicate rows

returns.drop_duplicates(inplace=True)

4.handling missing values

returns.isnull().sum():checking null values

there is no null values

Stores dataset

1.Understand the Dataset:

stores.head()

stores.tail()

stores.info()

stores.describe()

stores.dtypes

stores.columns

2.standardizing columns

stores.columns = stores.columns.str.lower().str.replace(' ', '_')

3.Remove duplicate rows

stores.drop_duplicates(inplace=True)

4.handling missing values

stores.isnull().sum():checking null values

there is no null values